

## About Me

Research scientist and engineer with 4+ years of experience in recommender systems, with accepted papers at KDD, SIGIR, and RecSys. Currently a Senior Research Scientist at AI VK. B.Sc. from Lomonosov Moscow State University; M.Sc. candidate at the Higher School of Economics.

## Research Interests

Information retrieval, conversational recommendation, large language models, generative recommendation, in-text reasoning for recommendation, reinforcement learning, and the mathematical foundations of deep learning.

## Work Experience

Mar 2026 – Present **Senior Research Scientist, AI VK**

Research team led by [Prof. Dr. Alexander D'yakonov](#), formerly world's #1 rank on Kaggle.

- Worked on long-term RL optimization for generative retrieval in recommender systems; first author of *Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE* (KDD 2026).
- Made major contributions to the internal research framework, speeding up experiment iteration several-fold.

Jul 2025 – Mar 2026 **RecSys R&D Team Lead / Senior Research Engineer, Yandex**

- Led a three-person R&D team responsible for pretraining generative recommendation models, internal training framework development, and inference optimization.
- Worked on pretraining a OneRec-style generative recommender that replaces the entire cascade pipeline (candidate generation, ranking, and re-ranking) with a single end-to-end neural network.
- Introduced FSDP (Fully Sharded Data Parallel) training, scaling recommender transformers up to 7 billion parameters with improved model quality.
- Built the training framework and infrastructure for large-scale generative recommender models.
- Worked on adapting sub-quadratic attention mechanisms for generative retrieval in recommender systems ([paper](#)).

May 2023 – Jul 2025 **RecSys Research Engineer, Yandex**

- Investigated graph neural networks for item representation in recommender systems ([talk](#)).
- Developed a transformer-based candidate generation model for Yandex Market and conducted A/B tests.
- Co-developed [Argus](#), a billion-parameter generative recommender transformer; achieved +2.26% Total Listening Time and +6.37% item like rate at Yandex Music in online A/B tests; later adapted across Yandex products (e.g., e-commerce +1.7% GMV, food-tech +2.1% GMV).
- Contributed to Argus adaptation for Yandex Ads ranking, delivering +2% revenue uplift in online A/B tests ([talk](#)).
- Mentored two interns, who have since become a Senior RecSys Researcher and a core developer at YandexGPT.

Feb 2022 – Jul 2022 **Backend Developer Intern, Yandex**

- C++ backend development: optimized runtime performance of search algorithms and data-preparation pipelines.
- Implemented new ML features and conducted A/B tests.

## Education

Sep 2024 – Present **Higher School of Economics (HSE)**, Moscow, Russia

Master's Program in Modern Computer Science.

Thesis: *Long-Term Optimization for Large-Scale Generative Retrieval*.

Sep 2024 – Present **Yandex School of Data Analysis (YSDA)**, Moscow, Russia

Selective postgraduate program in machine learning, deep learning, and applied mathematics.

Sep 2020 – Jun 2024 **Lomonosov Moscow State University (MSU)**, Faculty of CMC, Moscow, Russia

B.Sc. in Applied Mathematics and Computer Science.

Thesis: *Graph Neural Networks for Candidate Retrieval in E-Commerce*.

## — Selected Publications

- KDD 2026** [Long-Term Optimization for Large-Scale Generative Retrieval with Off-Policy REINFORCE](#), *5th Workshop on End-to-End Customer Journey Optimization*  
**Artem Matveev**, Sergei Makeev, Aleksei Krasilnikov, Vladimir Baikalov, Sergei Liamaev, Kirill Khrylchenko
- Preprint 2026** [Embedding Items at Scale: Comparing GNN-Based and ID-Based Item Embeddings in the Yandex Ecosystem](#)  
Sergei Makeev, **Artem Matveev**, Vladimir Baikalov, Kirill Khrylchenko  
[Scaling Recommender Transformers to One Billion Parameters](#)  
Kirill Khrylchenko, **Artem Matveev**, Sergei Makeev, Vladimir Baikalov
- SIGIR 2026** [Gated Bidirectional Linear Attention for Generative Retrieval](#)  
**Artem Matveev**, Vladislav Tytskiy, Sergei Makeev, Sergei Liamaev
- RecSys 2025** [Correcting the LogQ Correction: Revisiting Sampled Softmax for Large-Scale Retrieval](#)  
Kirill Khrylchenko, Vladimir Baikalov, Sergei Makeev, **Artem Matveev**, Sergei Liamaev

## — Teaching

- 2024 – 2026 Teaching Assistant, Yandex School of Data Analysis (YSDA)  
[DeepRecSys Course](#). GitHub (140+ stars).
  - Seminars on evaluation metrics, neural RecSys framework design, neural ranking, and reinforcement learning.
  - Authored homework assignments for the course.
- 2025 – 2026 Teaching Assistant, Higher School of Economics  
[Recommender Systems Course](#). GitHub (90+ stars).
- 2025 Teaching Assistant, Higher School of Economics  
[Deep Learning Course](#). GitHub (100+ stars).
  - Seminar on neural ranking.

## — Talks

- 2025 *Do We Need Graph Neural Networks to Encode Items in Recommendations?*, Yandex For ML  
[YouTube video](#).

## — Competitions

- 2023 [Kaggle “ICR — Identifying Age-Related Conditions”](#)  
Silver Medalist.
- 2022 [ACM ICPC, 2022 ICPC Northern Eurasia Finals](#)  
Diploma of the 3rd degree.
- 2020 [ACM ICPC, Final of Northern Eurasia 2020](#)  
Diploma of the 2nd degree.

## — Technical Skills

- Languages Current: Python, Triton, CUDA. Past: C++, Assembly.
- Frameworks PyTorch.
- ML Engineering Full production cycle: data preparation, experimentation, model export & conversion, performance benchmarking, Triton Inference Server deployment, real-time history processing.

## — Professional Service

- 2026 UMAP 2026 — PC member, Industry Track

## — Patents

- Dec 2025 Application for invention [2025136959](#), Federal Service for Intellectual Property  
Yandex filed a patent application for Argus (see [Scaling Recommender Transformers to One Billion Parameters](#)).